

Ch3

Spam, botnet, social media & teenagers

是什么？做了什么？为什么要这么做？好处/坏处？影响

Spam & Botnet

*What is spam?

Spam is defined as **unsolicited, bulk email**.

垃圾邮件被定义为未经请求的大量电子邮件

为什么要 send spam?

Spam is profitable – cheaper than "junk mail"

相比于传统的垃圾邮件（纸质邮件）

影响？

Consumed a large percentage of the Internet's **bandwidth** and huge amounts of **storage space on mail servers and individual computers**

消耗了互联网的大量带宽，占用了邮件服务器和个人计算机的巨大存储空间

*How firms get email addresses?

- Web sites, chat-room conversations, newsgroups

用户在网站注册、参与聊天室对话或加入新闻组时，可能会提供电子邮件地址

- Computer viruses

病毒和恶意软件可以从受感染的计算机中窃取电子邮件地址。

- Dictionary attacks

使用自动生成的常见用户名和域名组合来猜测有效的电子邮件地址

- Contests

使用自动生成的常见用户名和域名组合来猜测有效的电子邮件地址

How to send spam?

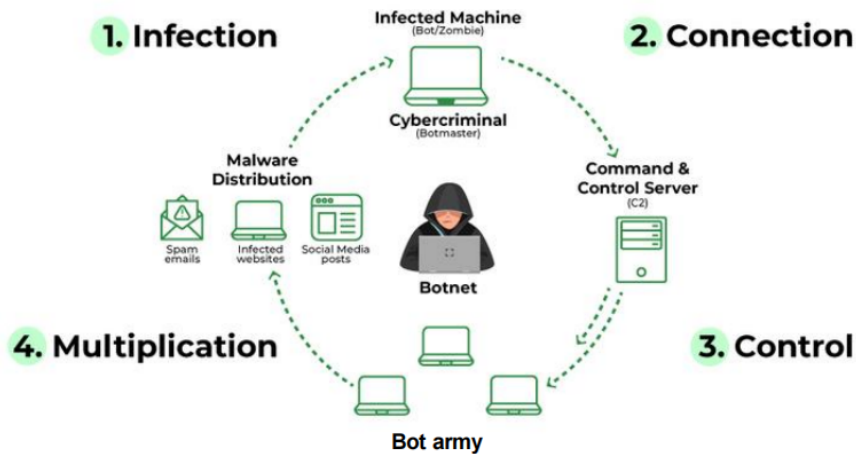
Most spam sent by bot herders (bot masters) who control **botnets**

***Botnets: huge networks of compromised computers (what?)**

大多数垃圾邮件是由“僵尸牧羊人” (bot herders 或 bot masters) 控制的僵尸网络 (botnets) 发送的。

僵尸网络：由大量受感染的计算机组成，这些计算机在不知情的情况下被用来发送垃圾邮件

How botnet works?



***How to fight spam?**

ISPs and businesses have installed **spam filters** 垃圾邮件过滤器

Law enforcement agencies around world making effort to shut down **botnet** command centers –全球的执法机构正在努力关闭僵尸网络 的指挥中心。

Social–technological cycle&spam

- Email messages practically free
- Profits increase with number of messages sent
- Strong motivation to send more messages
- Design of Internet allows unfair, one-way communications
 - Spammers disguise their own email addresses

Social media & teenagers

Two important attributes to make Web a global tool for information exchange:

- It is **decentralized**, everyone can add new information
- Every Web object has a **unique** address (the URL)

What is the relationship between?

Influence

The rates of depression and anxiety and suicide 自杀 and loneliness going up among people. Social media is an important driver of that youth mental health crisis.

Pron and Cons?

Pros

1. Anyone can publish their thoughts online
2. Free source of information
3. Help us make friends
4. Allow for dissent 不同意
5. Important educational tools

Cons

1. Lead to addiction
2. Increase online bullying
3. Lead to body image issues
4. Spread misinformation
5. Cause sleeplessness
6. Increase stress

Web filters

What?

Software that prevents display of certain Web pages

Who provide the service?

- May be installed on an individual PC
- ISP may provide service for customers

How? (Methodologies) –No foolproof

- Maintain "black list" of objectionable 有异议的 sites
- Examine content for objectionable words/phrases

Child Internet Protection Act (CIPA)

- Libraries receiving federal networking funds must filter pages containing obscenity or child pornography

接受联邦网络基金资助的图书馆必须过滤含有淫秽或儿童色情内容的网页

- The law allows adults who desire access to a blocked page to ask a librarian to remove the filter

该法律允许想要访问被屏蔽页面的成年人要求图书管理员移除过滤器

- U.S. Supreme Court ruled CIPA did not violate 1st Amendment guarantees.

美国最高法院裁定CIPA没有违反第一修正案的保证。

Sexting

What?

Sexually suggestive 暗示的 text messages or emails with nude or nearly nude photographs

Influence

Heavily impacting (other) people's lives